

Cloning

Storage of cells

- The requirement for continued storage of cells at all stages is essential for the following reasons:
 - ★ Throughout cloning, particularly during the early stages, there is a high probability of chromosome loss and consequent loss of Ig secretion from the hybridoma cells.
 - ★ There is always a risk of overgrowth by Ig-nonproducing hybridomas
 - ★ The risk of infection.

Cloning

- The purpose of cloning is to isolate a single-specific Ab-secreting cell from the hundreds of other cells in the culture. So that Monoclonal cell line can be established.
- The types of hybridoma cells depends on the number of epitopes of the injected Ag.
- If the Ag contain 5 epitopes → 5 different types of Hybridoma cells each secrete Mab.

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- Fetal calve serum must be test before used to support Hybridoma growth.
 - Small sample of Hybridomas seeded in media (serum was added to media).
 - After five days, if the cells growth is supported use it, if not change it.



- Hybrids are seeded into culture wells containing such cells.

- ★thymus/96-well plate +wash medium + 10 cells /100 uL/well (12 wells or 1 row)

- ★5% FCS (to be tested) in medium + -ve and +ve controls

- ★After 5 days score each well for hybrid growth

- Usually, the growth of hybrids is inhibited at high cell dilutions. This can be overcome by the use of feeder cells such as thymocytes, peritoneal macrophages or spleen cells.

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- Immunization Ag contain 5 epitopes Isolation of Plasma cells→specific Plasma cells (Against target Ag) and non specific plasma cells (Against other Antigens) →fusion with Myeloma cells →selection→1000 hybridoma cells→ 5 Types of Hybridoma cells specific (Produce Mab against target Antigen) other types non specific (Against other Antigens)→ each cell isolate alone, and allow to propagate to produce clone of cells each recognize the same epitope.

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- Each clone must screen by ELISA :
 - **Wells that give positive result:** specific Hybridoma cells
 - **Wells that give negative result:** non specific Hybridoma cells.
 - Maybe thousands of Hybridoma cells result from fusion how we can isolate each cell alone?
 - Cloning by limiting dilution

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- The objective is to set up at least 3 very low dilutions of cells resulting in a mean of 1 cell/well.
 - Within 2 weeks of cloning, clones should be visible macroscopically and some of the wells contain monoclonal growth.
 - At this stage, each of the cell supernatant should be screened to determine which of them contains the desired Ab.
 - Positive wells are expanded and recloned at least 3 times
 - The final clone should display positive Ab production in all the seeded wells.
 - A proportion of each culture should be frozen and stored.

Limiting dilution

1. Hybridoma cells mixture
2. Count them the number of cells.
3. Make dilution to 2000 cell/ml = diluted Hyb cells
4. Add 100 uL media to every 96 well
5. Pipett 100 uL of diluted Hybridoma cell to A1 well then
6. serial dilution horizontally to A12
7. Make serial dilution to each well starting from column
8. A1 to H1, A2 to H2 etc.....
9. Let the cells grow
10. ELISA screening
11. Repeat the same dilution procedure for every positive clone

Alternative limiting dilution

1. 10 tubes

2. Tube #1 contains 10,000 cells/ml

3. Tube #2 contains 4.5 ml media

4. Tube #3 -5 contains 4 ml media

5. Tube 6-15 contains 100 μ L media



6. Transfer 0.5 ml from Tube 1 to 2.....1000
cells/ml 1:10 dilution

7. Transfer 1ml from tube 2 to 3.....200 cells /ml
1:5 dilution

8. Transfer 1 ml from 3 to 4..... 40 cell/ml
1:5 dilution

9. Transfer 1 ml from 4 to 5 8 cells/ml
1:5 dilution

10. Transfer 100 uL from tube 5 to each of the
other tubes or wells

Alternative limiting dilution

1. Dilute the hybridoma cells to 2000 cells/ml
2. 100 μ L of these + 3.9 ml of thymocyte medium makes 4 ml
3. Plate 24 wells (row A & B) of 96-well plate 100 μ L each (5 cell/well)
4. To the 1.6 ml remaining, add 2.4 ml thymocyte medium



5. Plate 24 wells (row C & D) of 96-well plate
100 μ L each (2 cells/well)

6. To the 1.6 ml remaining, add 1.6 ml
thymocyte medium

7. Plate 24 wells (row E & F) of 96-well plate
100 μ L each (1 cells/well)

8. Expand the Mab-producing isolated
hybridoma clones, freeze and propagate

Semi solid hybridoma cloning

- Semi solid media: methylcellulose-based media
- Combine between Cloning and selection in one step.
- Reduces the overall time necessary to produce monoclonal cultures by up to 19 days.
- Semi solid media contain HAT media

Using the standard protocol for selection and cloning of hybridomas:

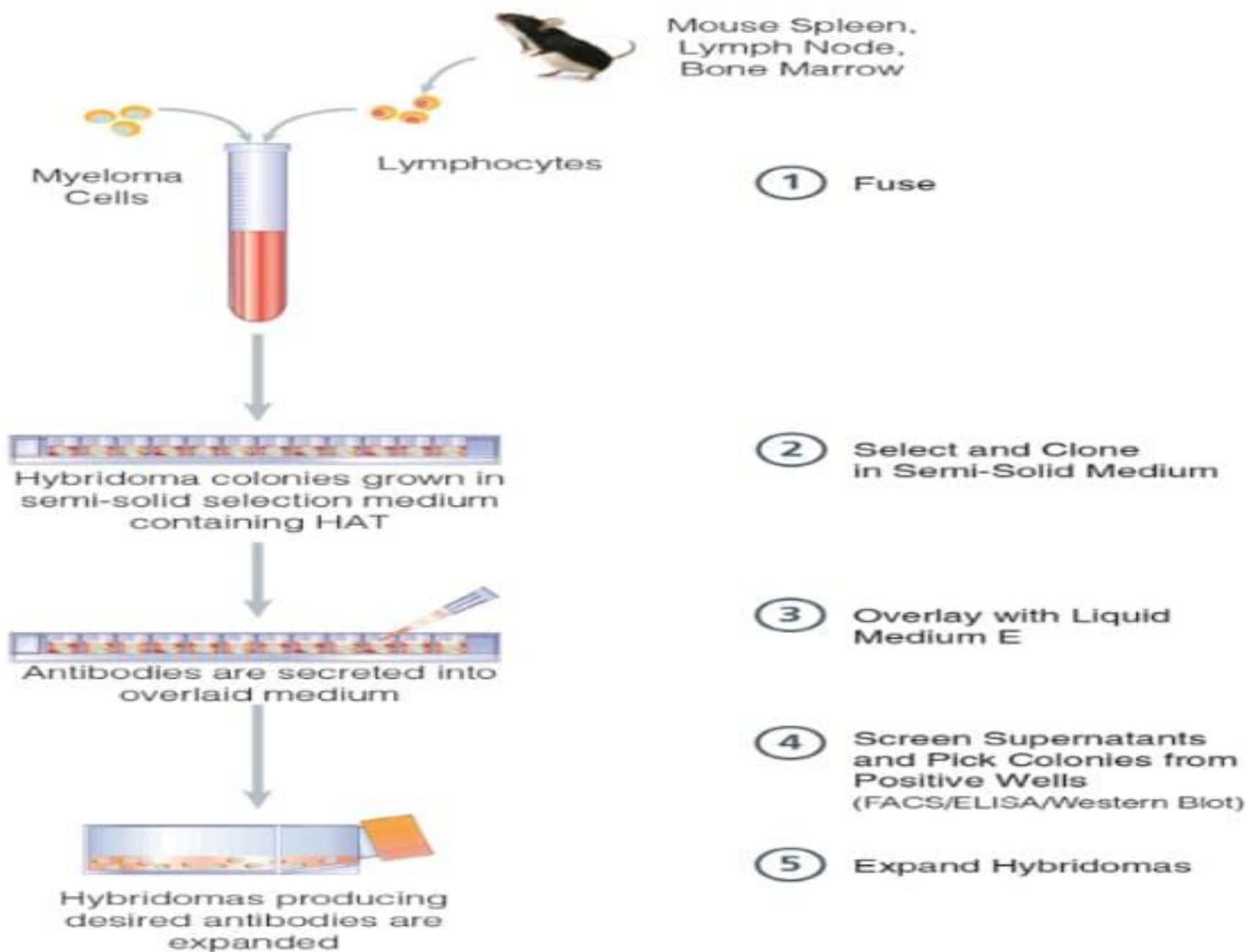
- 1.fused myeloma cells and splenocytes are suspended in selective semi-solid medium and incubated in 10 cm plates.
2. Individual cells grow to form discrete, monoclonal colonies that are picked from the semi-solid medium after 10 – 14 days of incubation
- 3.Colonies are then transferred to individual wells of a 96-well plate and cultured in liquid medium prior to screening the supernatants for clones of interest.
- 4.Each cell are picked in semi solid media, semi solid restrict the movement of cells, which mean cells belong to same clone will be close to gather, this make advantage over growing cells in liquid media.



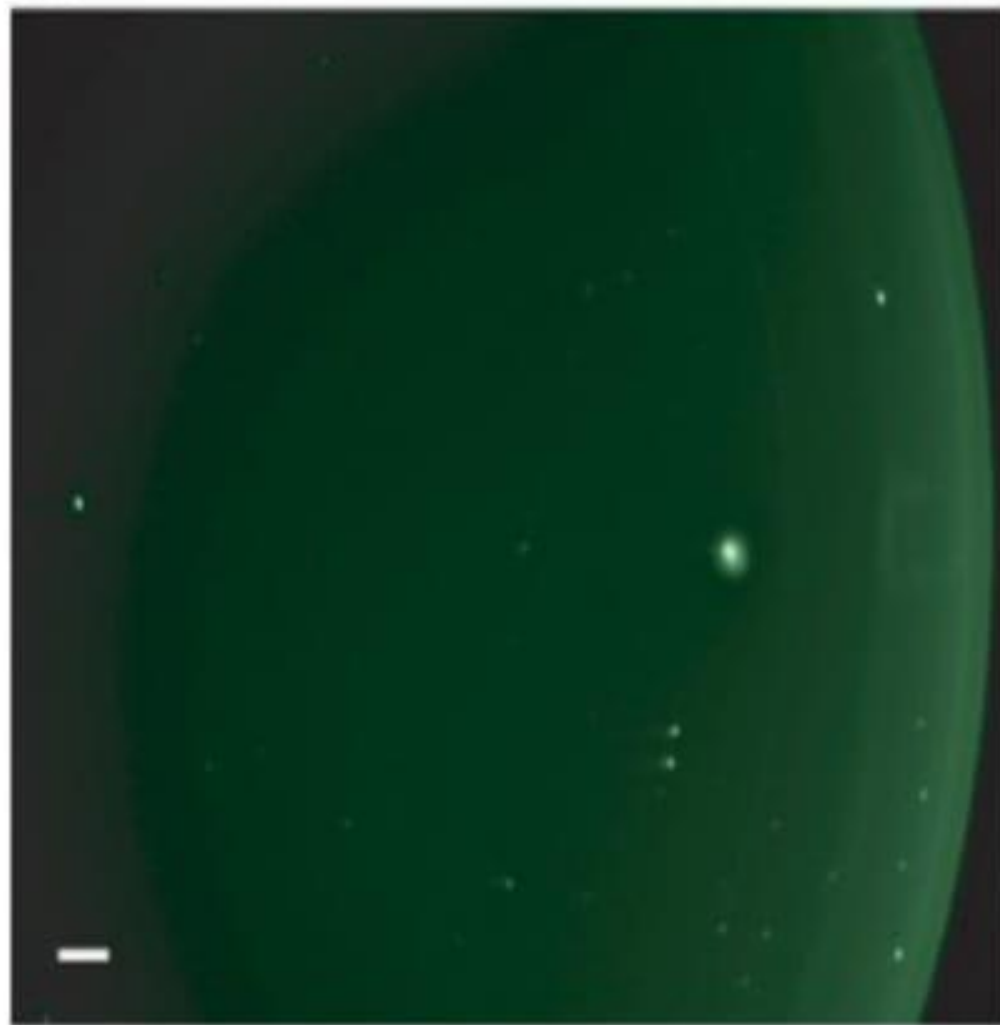
5.Only hybridoma cells will survive

6.In addition Semi solid media cover by Liquid media, Antibodies will Diffuse from Hybridoma cells to liquid media to allow screening for each colony to identify the specific colony that secrete the desired Mab.

- This method enables isolation of producing hybridomas with a high probability of monoclonality after only a single round of cloning. These combined benefits result in considerable time and labor savings.



a



b

